# Course Notes

## Lecture 10 — 6TiSCH and Routing in Low-Power Networks (RPL, CoRPL, CARP)

*Communication Technologies for the Internet of Things*

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*Text to support independent study*

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## How to Use These Notes

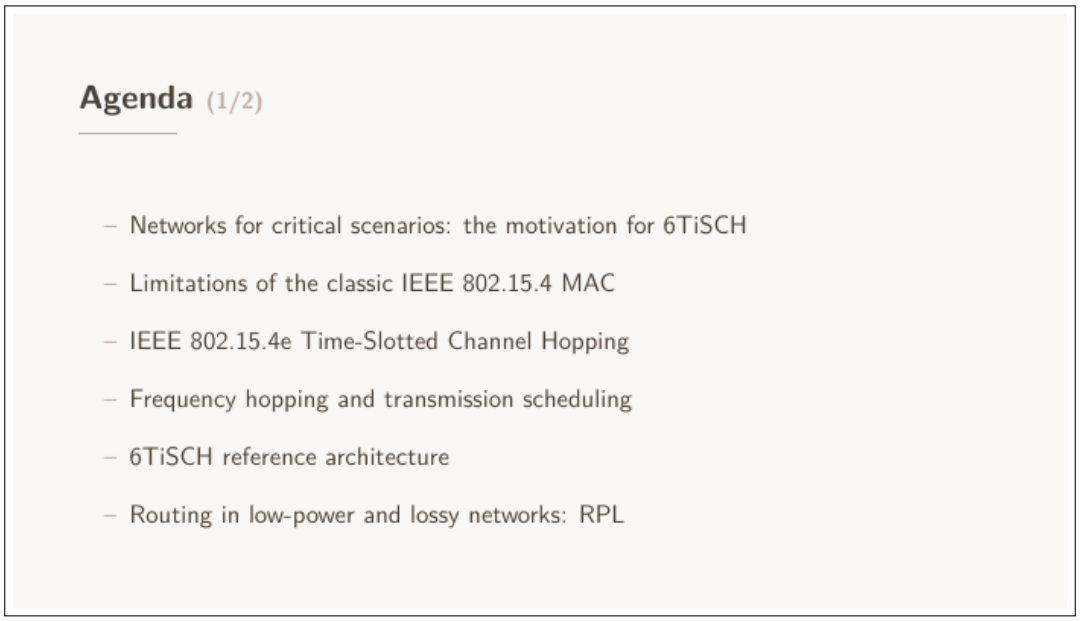
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

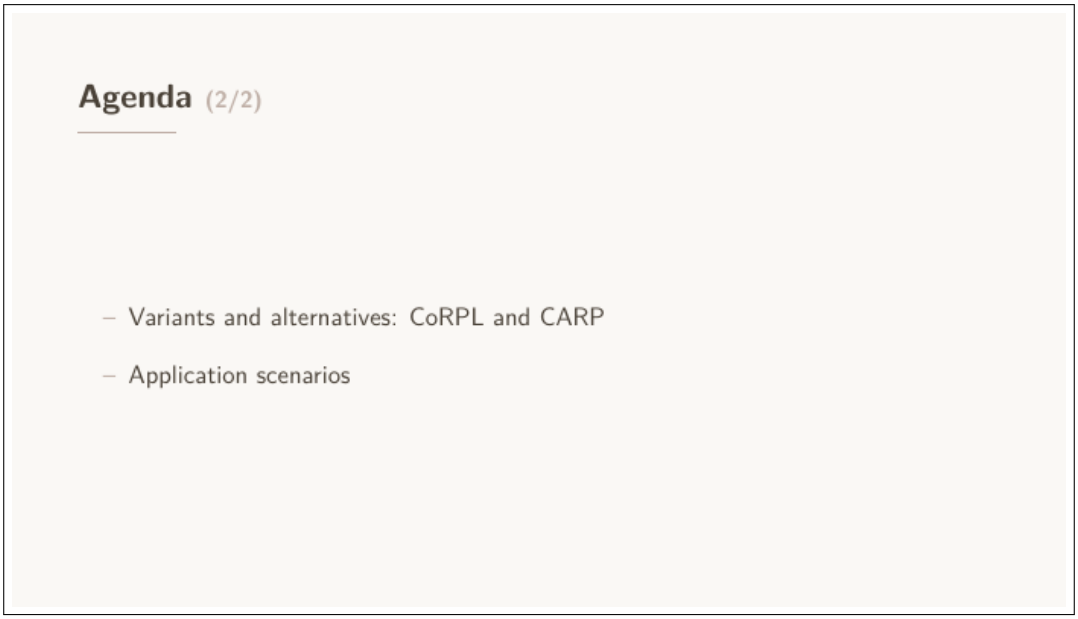
## 1. Commentary on the Slides

### Slide 2. Agenda (1/2)



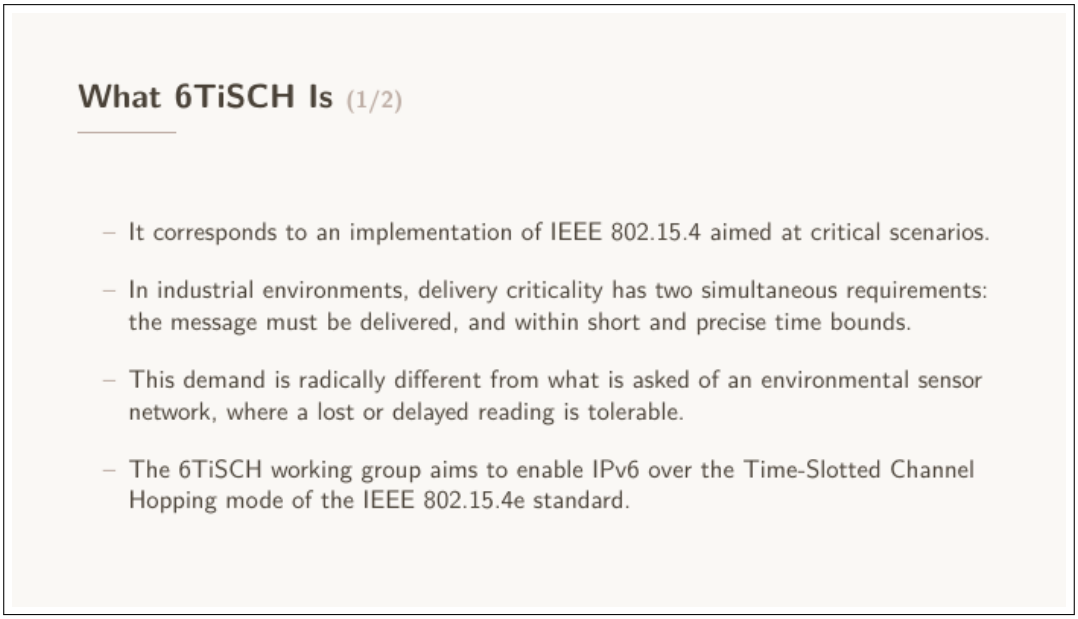

**The path through the lecture.** The lecture has two parts. The first deals with the link layer: why the classic 802.15.4 MAC does not serve critical scenarios and what TSCH does differently. The second deals with routing: how routes are built in a network of constrained devices, with RPL and its alternatives.

### Slide 3. Agenda (2/2)



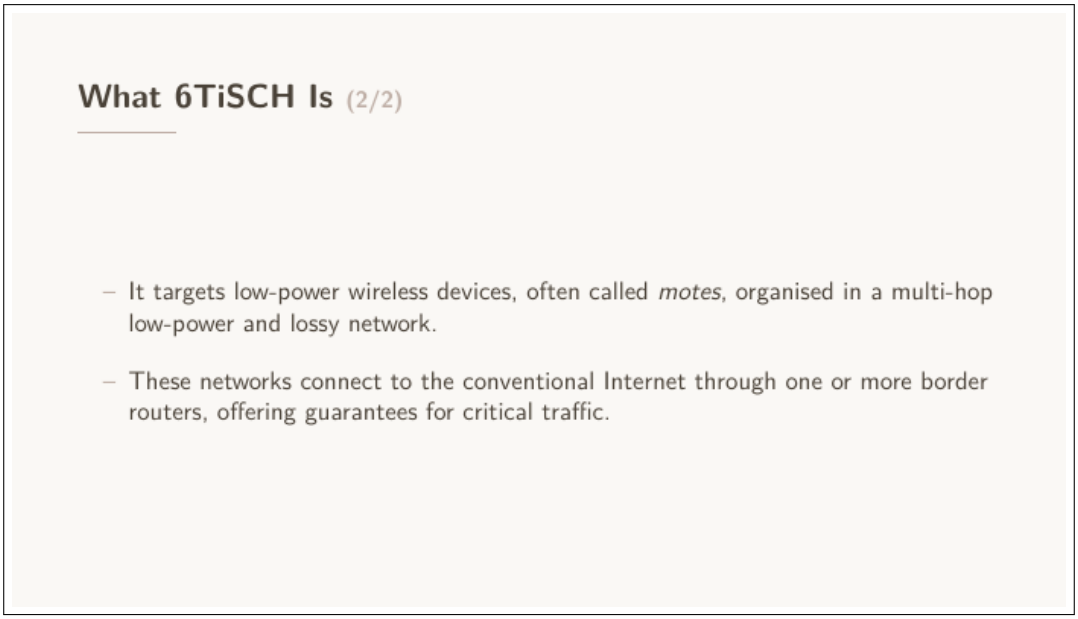

**Variants and scenarios.** CoRPL and CARP appear as alternatives to RPL for more unstable media. The application scenarios close the path and show where these technologies are actually used.

### Slide 4. What 6TiSCH Is (1/2)



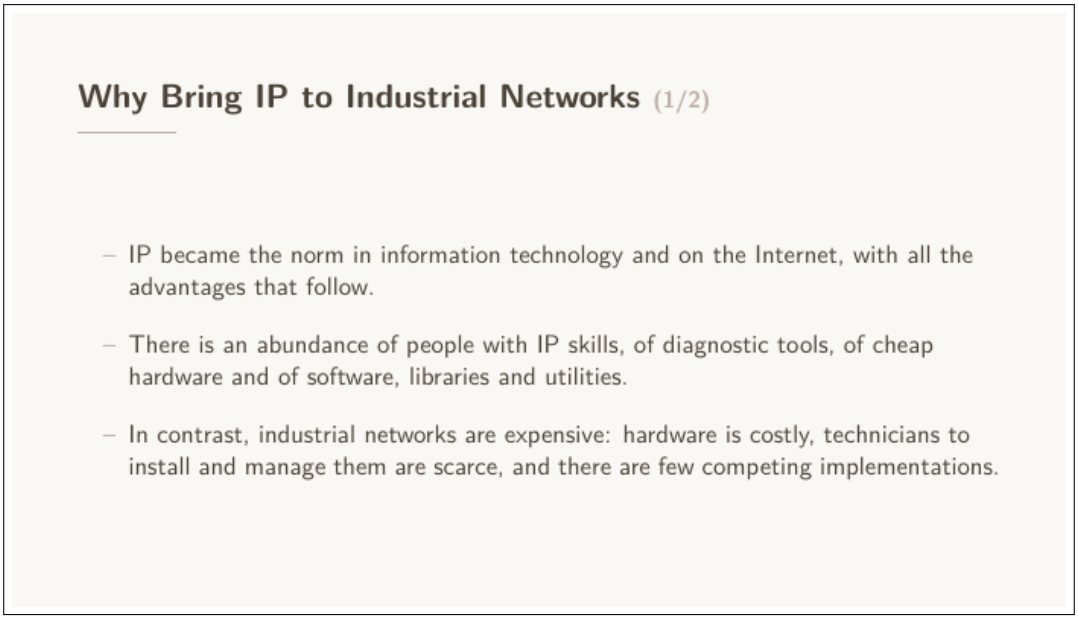

**A MAC for critical scenarios.** 6TiSCH puts IPv6 on the TSCH mode of 802.15.4e. The word critical has two parts: the message must arrive and it must arrive on time. In an environmental sensor network a lost reading is tolerable; in a factory it is not.

### Slide 5. What 6TiSCH Is (2/2)



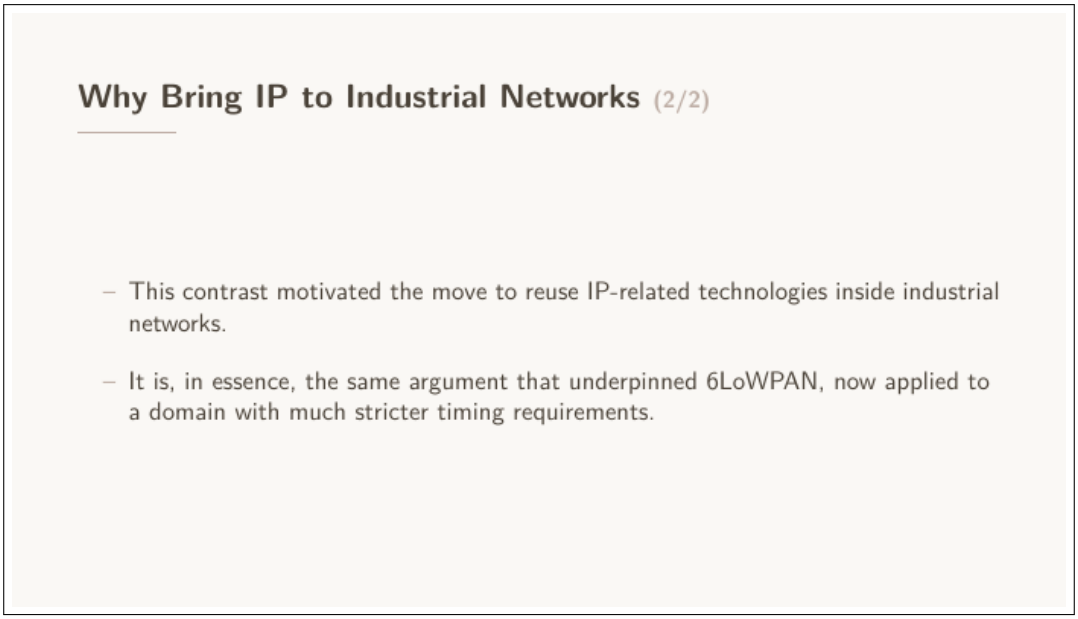

**Motes and border routers.** The devices are called motes and form a lossy multi-hop network. The connection to the Internet is made by one or more border routers, which offer guarantees to critical traffic.

### Slide 6. Why Bring IP to Industrial Networks (1/2)



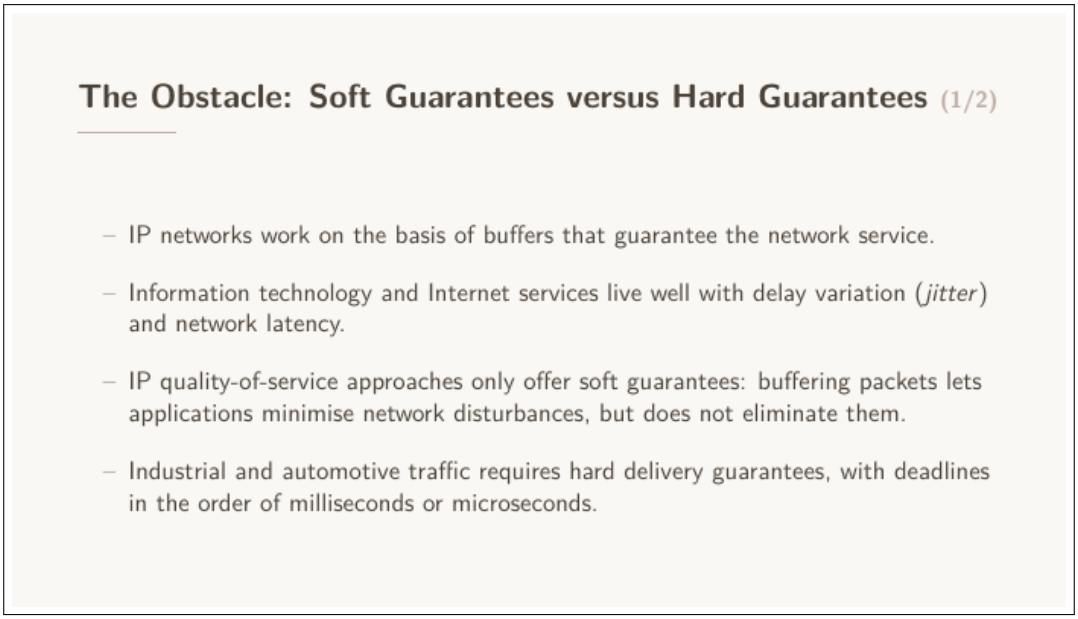

**The 6LoWPAN argument, again.** IP brings people, tools and cheap hardware. Traditional industrial networks are expensive and have few suppliers. It is an economic and skills argument, not just a technical one.

## Slide 7. Why Bring IP to Industrial Networks (2/2)



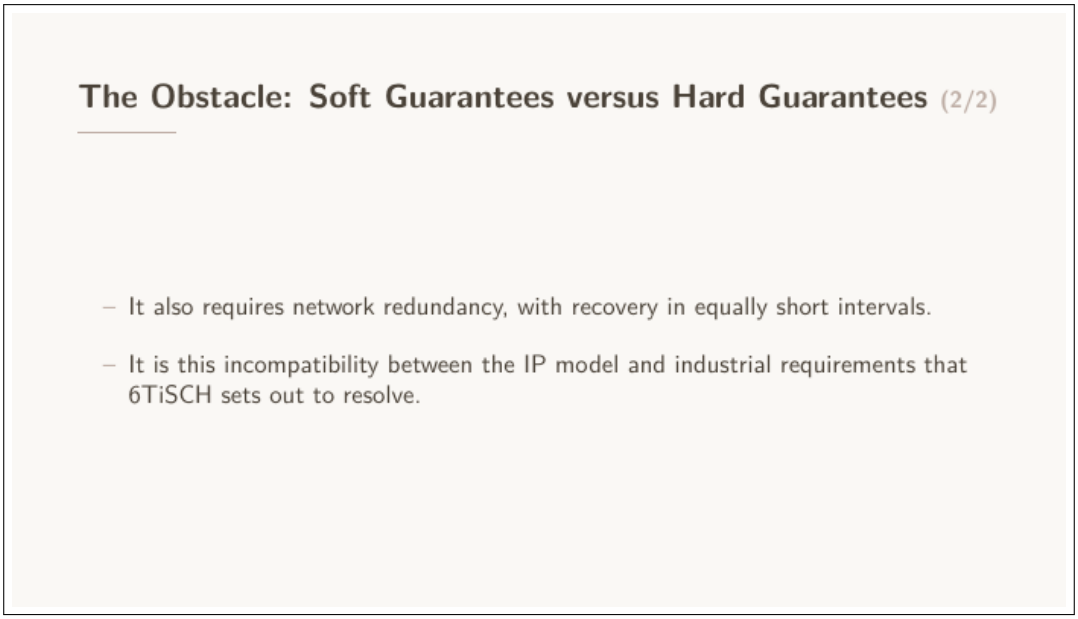

**Reusing IP in industry.** The motivation is the same as for 6LoWPAN, applied to a domain that is far more demanding in time. What changes is the price IP has to pay to meet it: determinism.

## Slide 8. The Obstacle: Soft Guarantees versus Hard Guarantees (1/2)



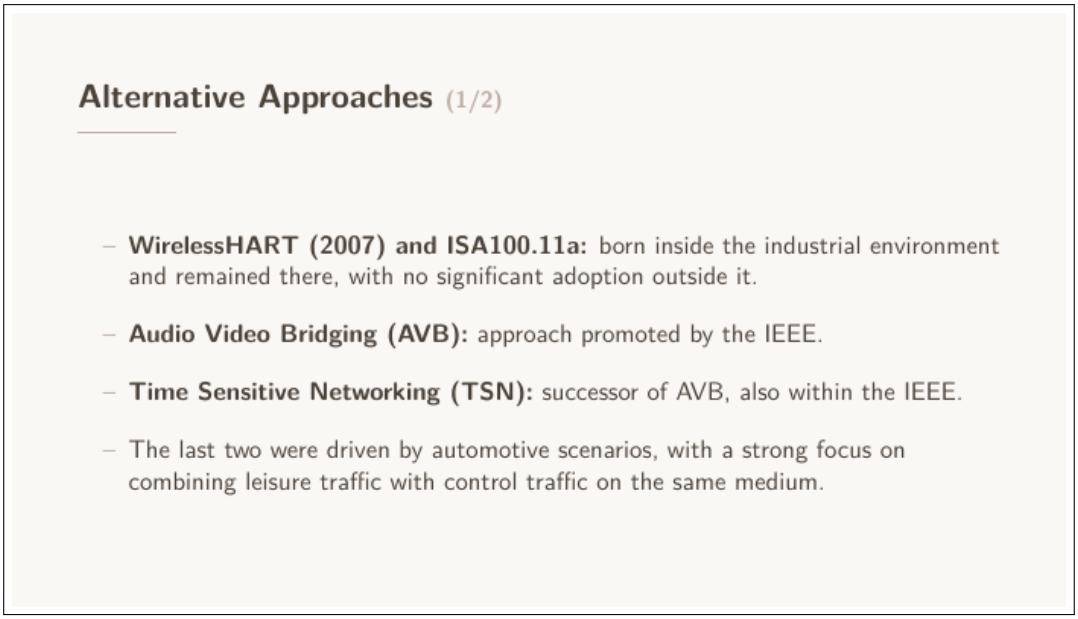

**Soft guarantees.** IP works well with delay variation because it uses buffers. Quality-of-service approaches only minimise the disturbance, without eliminating it. For a control application with a millisecond deadline, that is not enough.

### Slide 9. The Obstacle: Soft Guarantees versus Hard Guarantees (2/2)



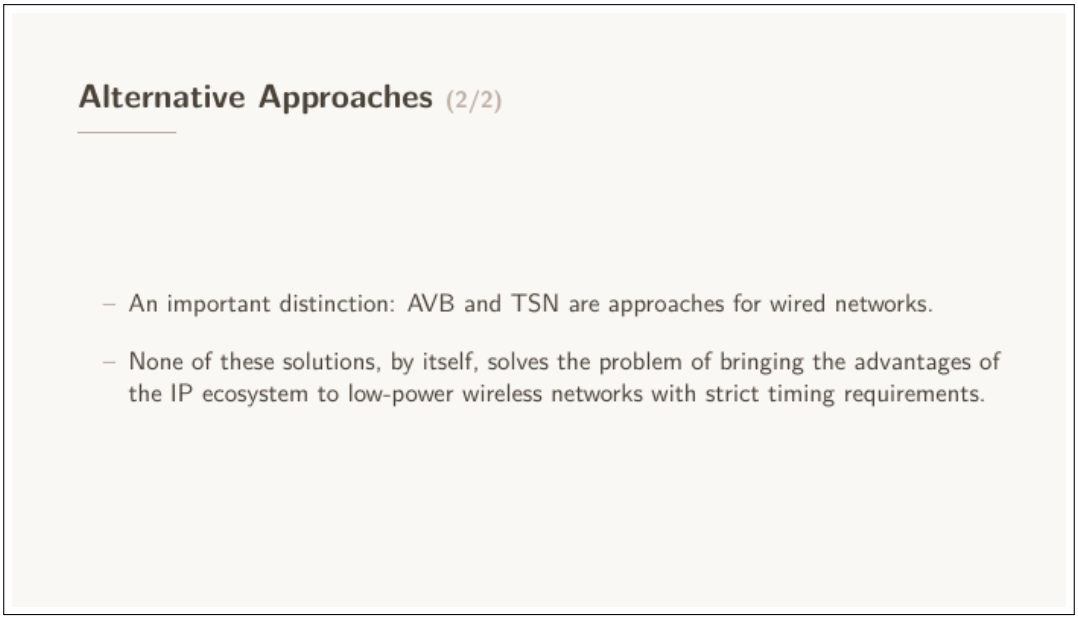

**Hard guarantees and redundancy.** Industrial traffic requires hard deadlines and redundancy with fast recovery. The incompatibility with the IP model is the problem 6TiSCH sets out to solve.

## Slide 10. Alternative Approaches (1/2)



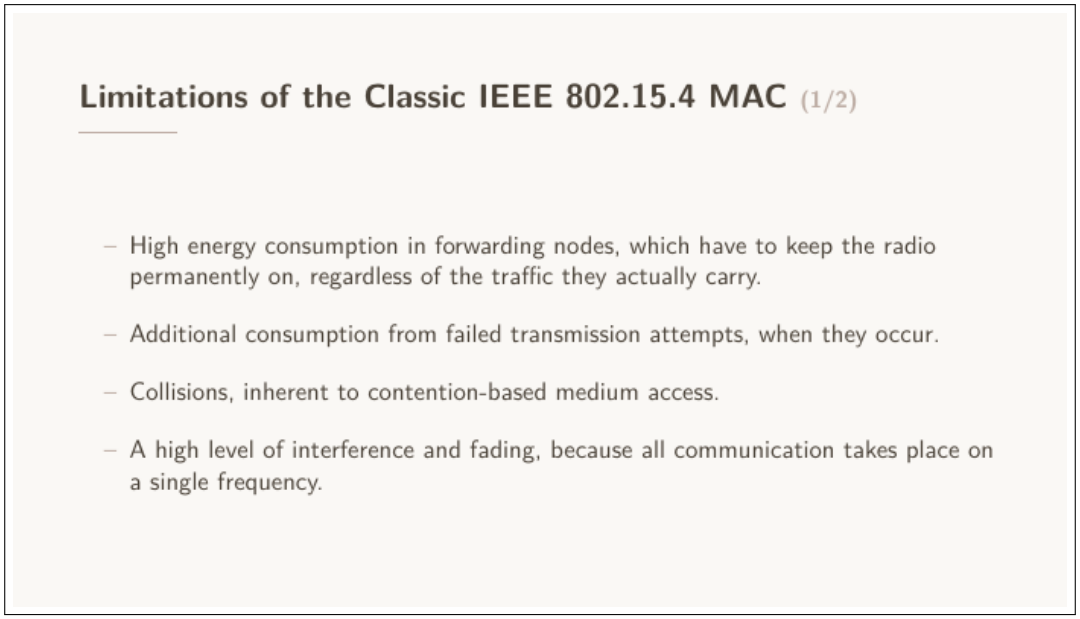

**Solutions that already existed.** WirelessHART and ISA100.11a were born in industry and stayed in industry. AVB and TSN come from the automotive world and are for wired networks. None combines IP, wireless and strict timing requirements.

## Slide 11. Alternative Approaches (2/2)



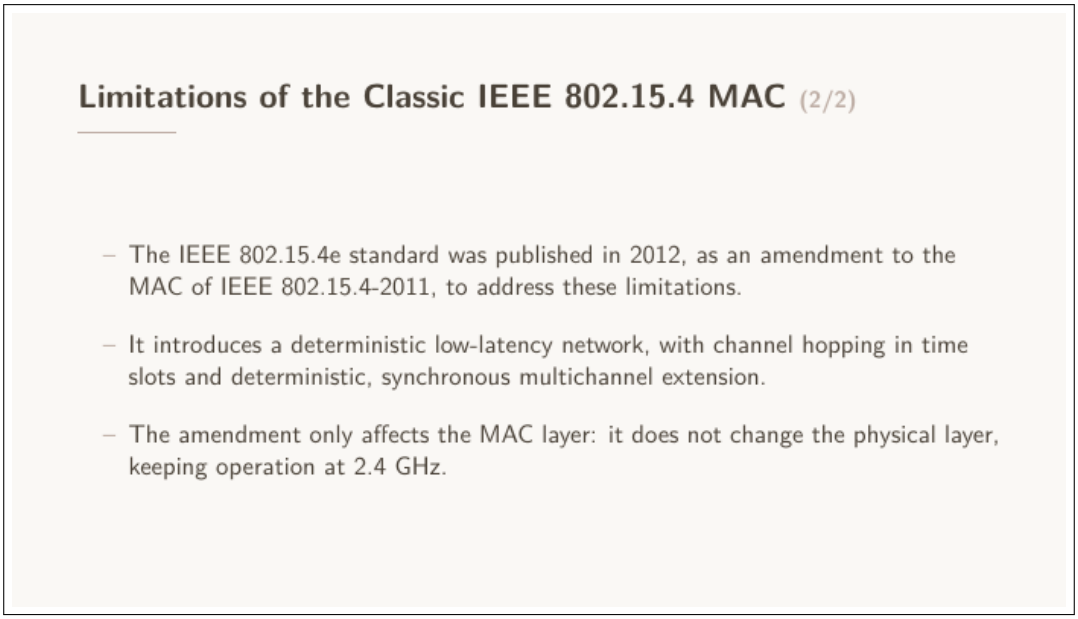

**What is missing.** The relevant distinction is the physical medium: AVB and TSN are wired. The gap 6TiSCH fills is the combination of IP, low-power radio and timing guarantees.

### Slide 12. Limitations of the Classic IEEE 802.15.4 MAC (1/2)



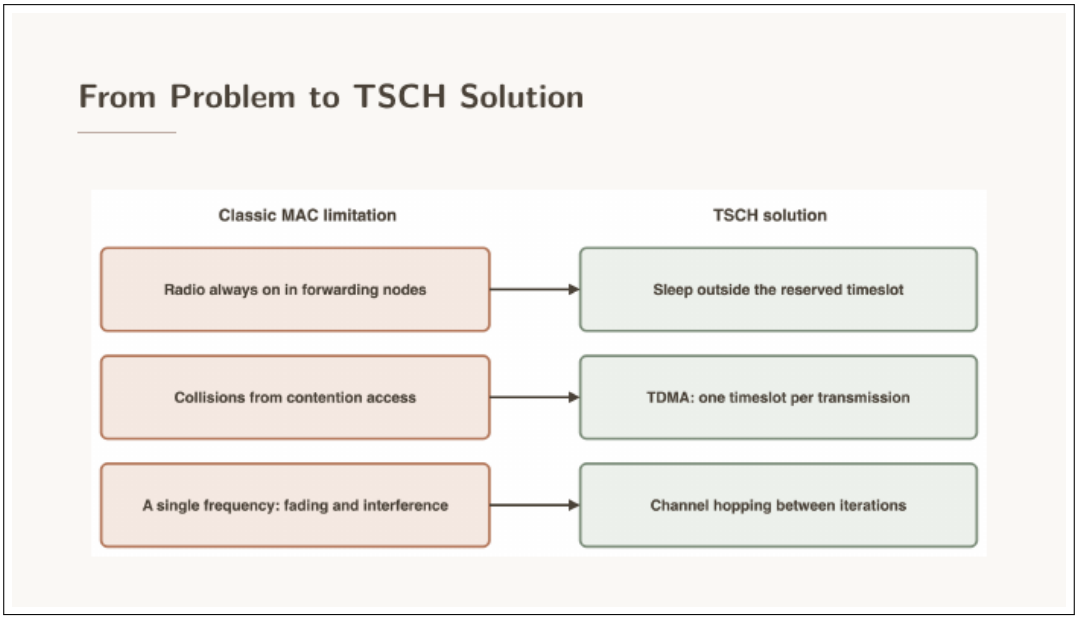

**Three defects of the classic MAC.** Radio always on in forwarding nodes, contention collisions and a single frequency exposed to interference and fading. Each has consequences for energy or reliability.

### Slide 13. Limitations of the Classic IEEE 802.15.4 MAC (2/2)



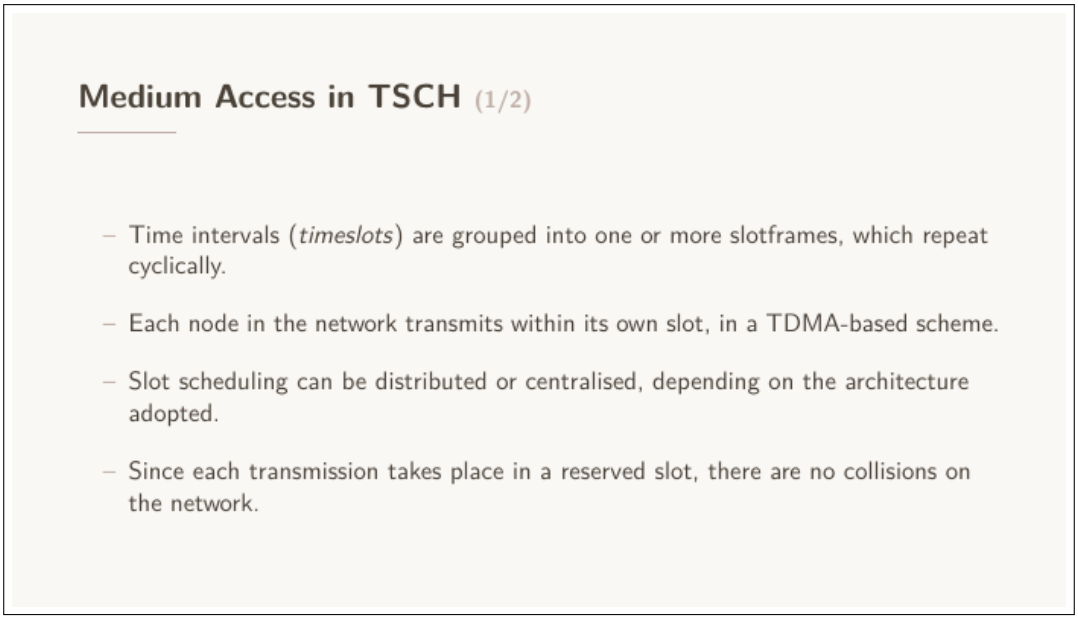

**The 802.15.4e amendment.** 802.15.4e changes only the MAC, and the physical layer stays at 2.4 GHz. This allows existing radio hardware to be reused with a new behaviour.

### Slide 14. From Problem to TSCH Solution



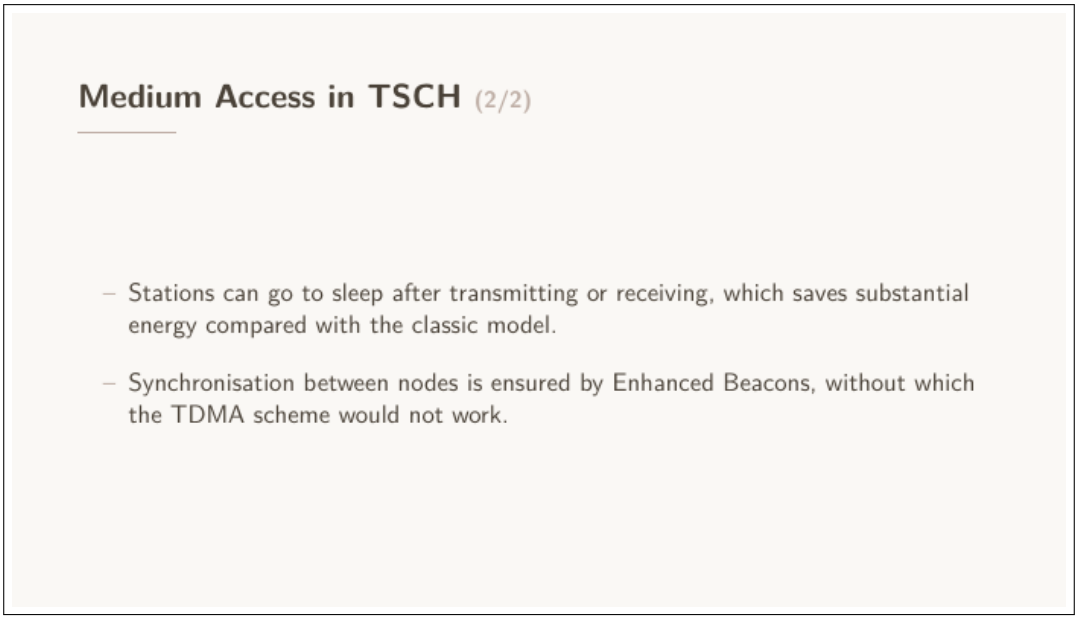

**From problem to solution.** The figure matches each defect of the classic MAC with TSCH's answer: sleeping outside the reserved slot, TDMA with one slot per transmission and channel hopping between iterations. It is a good summary to memorise.

### Slide 15. Medium Access in TSCH (1/2)



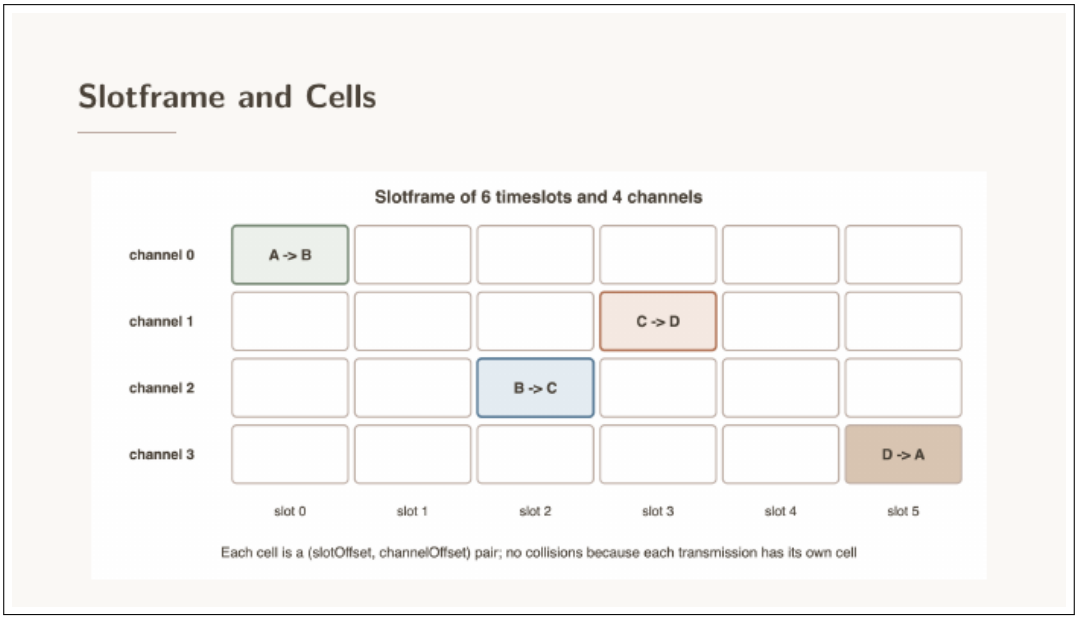

**TDMA with slotframes.** Slots are grouped into slotframes that repeat. Each transmission has its own slot, so there are no collisions. Scheduling can be central or distributed.

### Slide 16. Medium Access in TSCH (2/2)



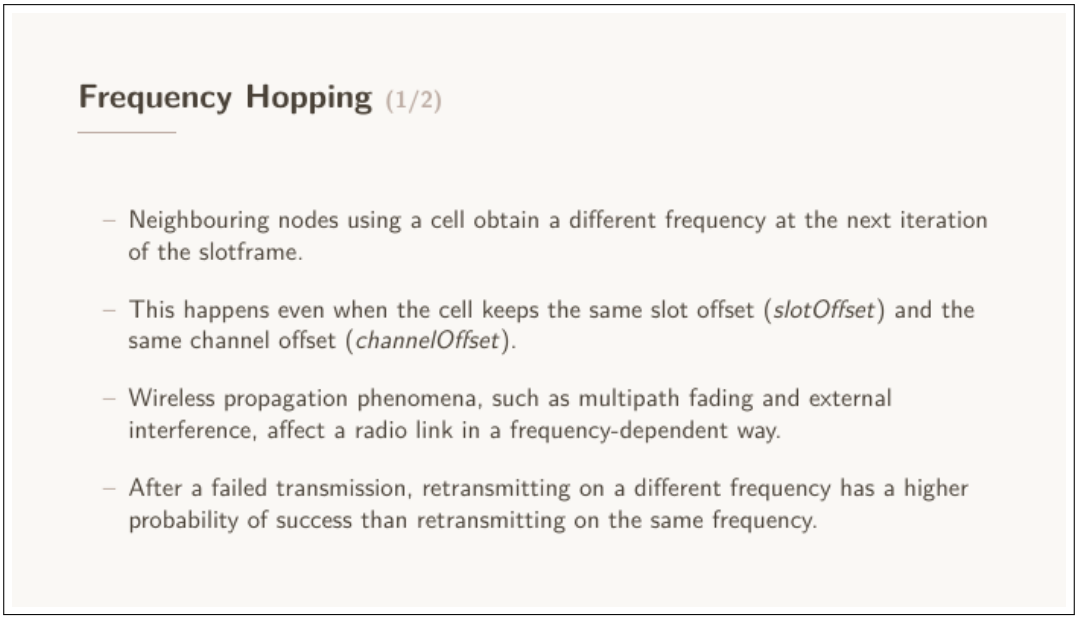

**Sleeping and synchronising.** Outside its slot, the radio sleeps, and that is where the energy saving comes from. Enhanced beacons keep clocks synchronised, and without them TDMA would not work.

## Slide 17. Slotframe and Cells



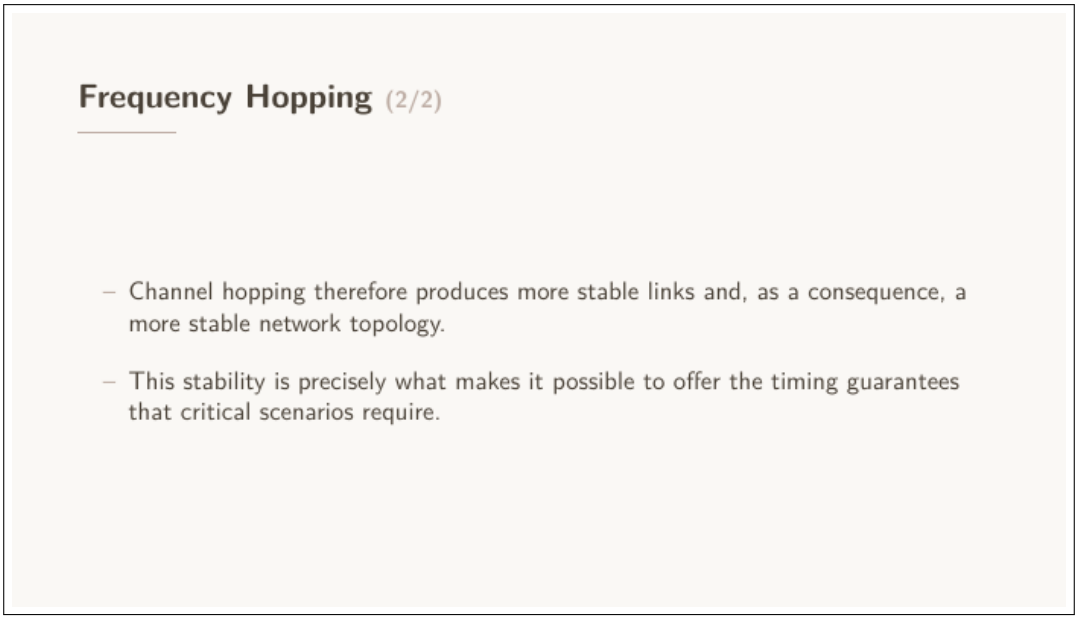

**The grid of cells.** Each cell is a slot and channel pair. The figure shows a slotframe of 6 slots and 4 channels, with four cells assigned to four links. Two links can use the same slot on different channels without colliding.

### Slide 18. Frequency Hopping (1/2)



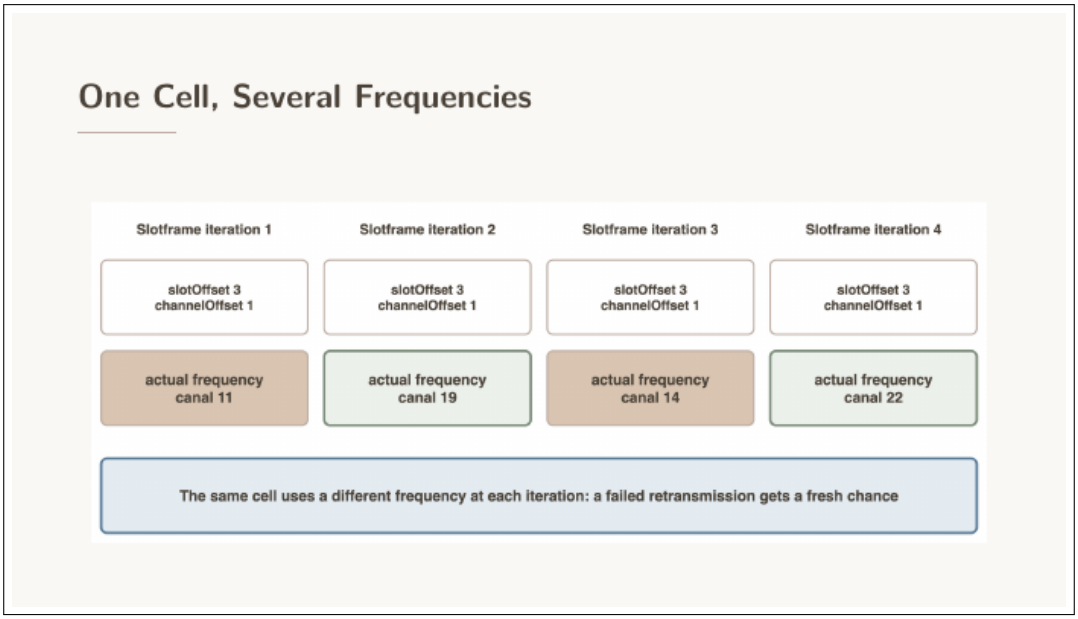

**Same cell, other frequency.** The same slot offset and channel offset correspond to a different actual frequency at each slotframe iteration. Since fading depends on frequency, changing frequency on a retransmission raises the probability of success.

### Slide 19. Frequency Hopping (2/2)



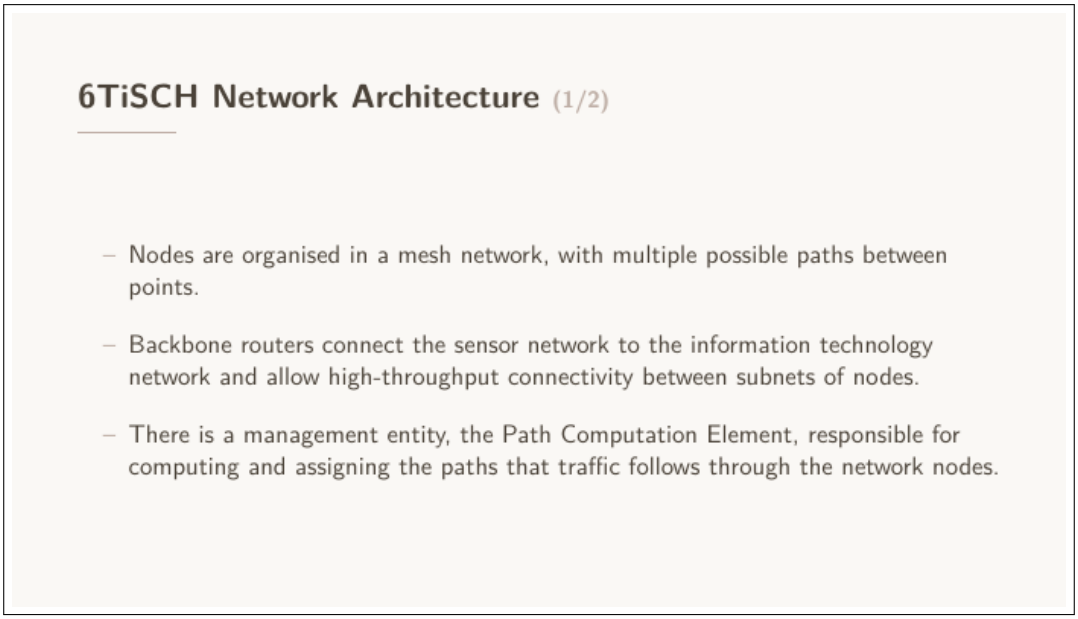

**Stable links.** Channel hopping produces more stable links and, as a consequence, a more stable topology. That stability is what makes it possible to offer timing guarantees.

## Slide 20. One Cell, Several Frequencies



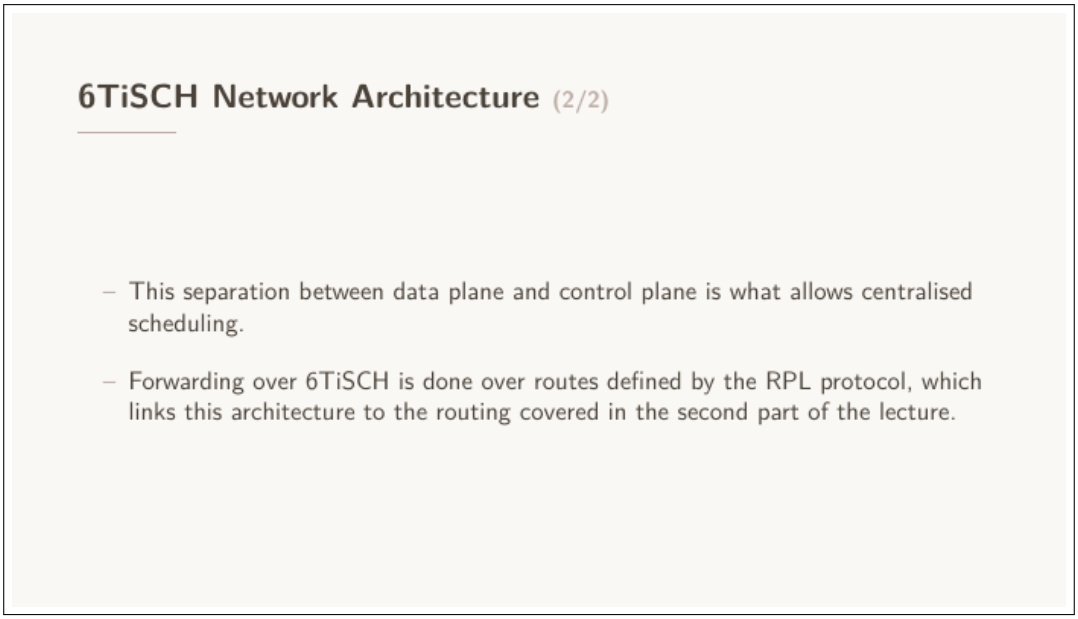

**Four iterations.** The figure shows the same cell, slotOffset 3 and channelOffset 1, landing on four different frequencies. It is the central idea of channel hopping in TSCH.

### Slide 21. 6TiSCH Network Architecture (1/2)



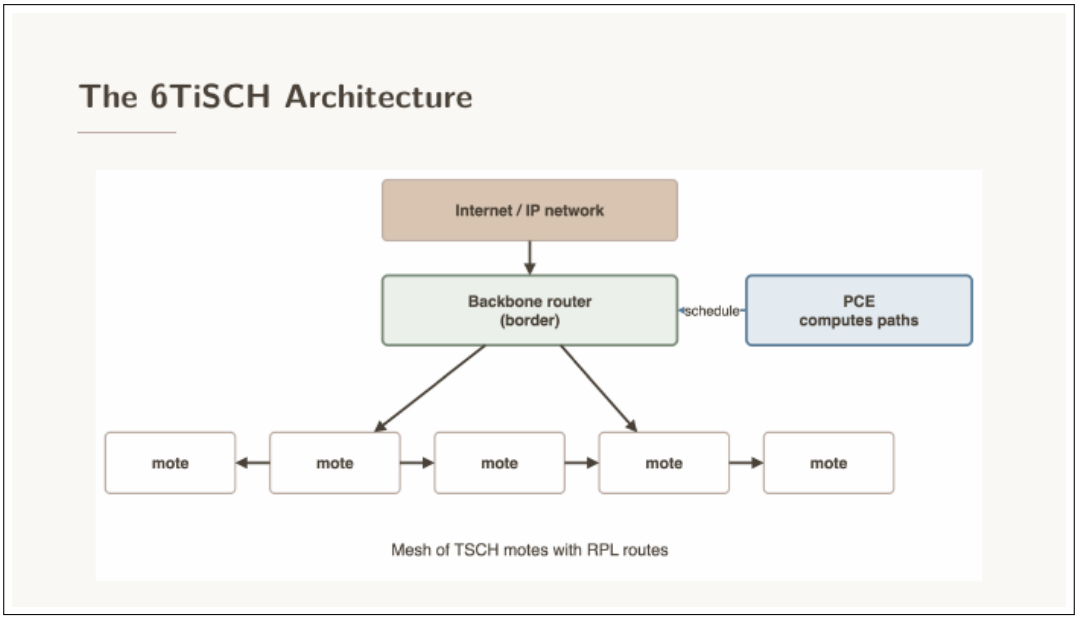

**Mesh, backbone and PCE.** The network is a mesh of motes with several paths. Backbone routers connect it to the IP network, and the Path Computation Element computes and assigns paths.

### Slide 22. 6TiSCH Network Architecture (2/2)



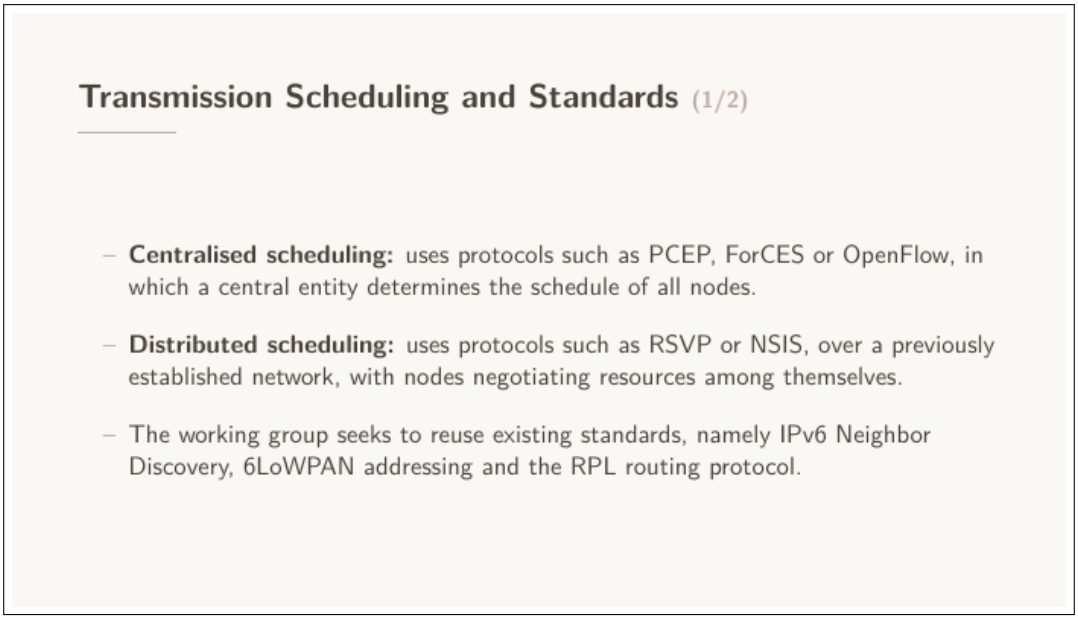

**A separate control plane.** Separating who computes from who forwards enables centralised scheduling. Forwarding itself is done with RPL routes, which links this part to the second half of the lecture.

### Slide 23. The 6TiSCH Architecture



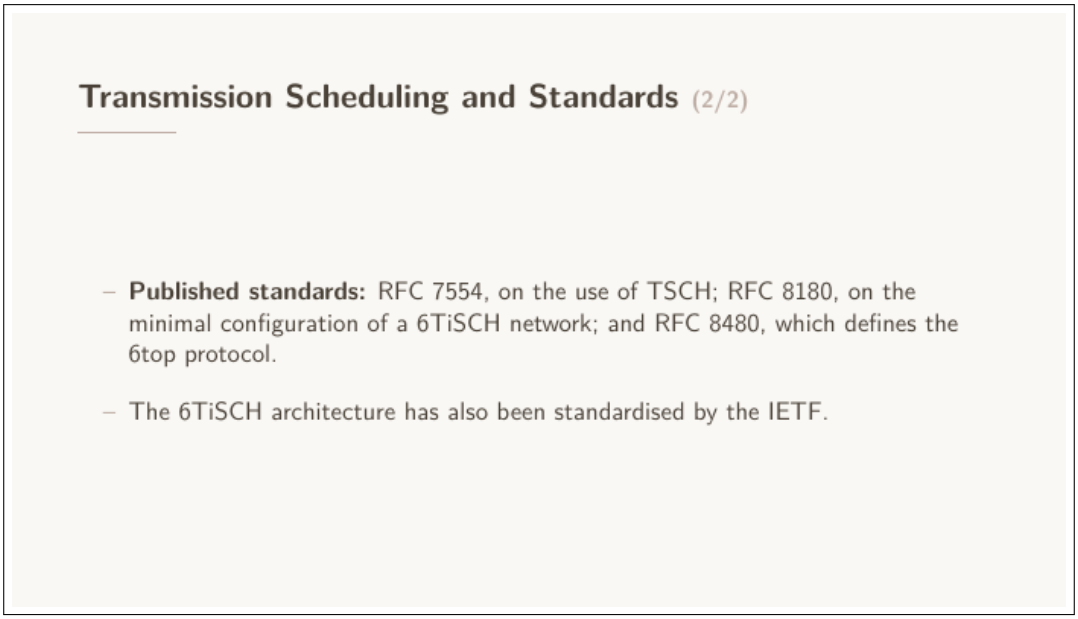

**The components of the architecture.** The figure shows the Internet, the border router, the PCE and the motes. The dashed arrow from the PCE to the router is the schedule that is distributed through the network.

### Slide 24. Transmission Scheduling and Standards (1/2)



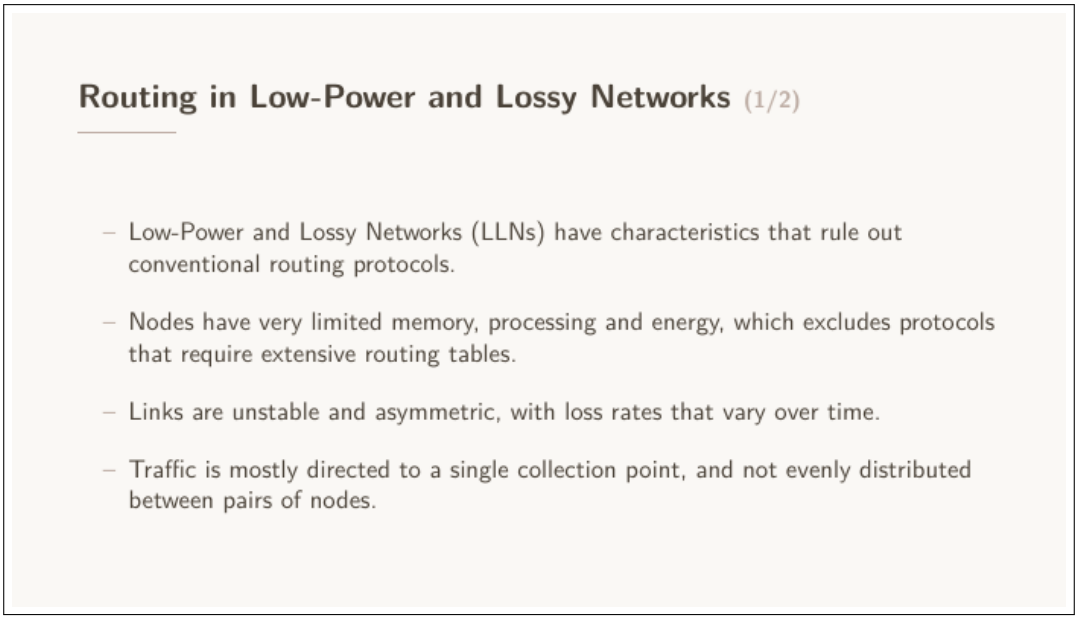

**Two ways to schedule.** Centralised, with PCEP, ForCES or OpenFlow, or distributed, with RSVP or NSIS. 6TiSCH tries to reuse existing standards instead of creating new ones.

### Slide 25. Transmission Scheduling and Standards (2/2)



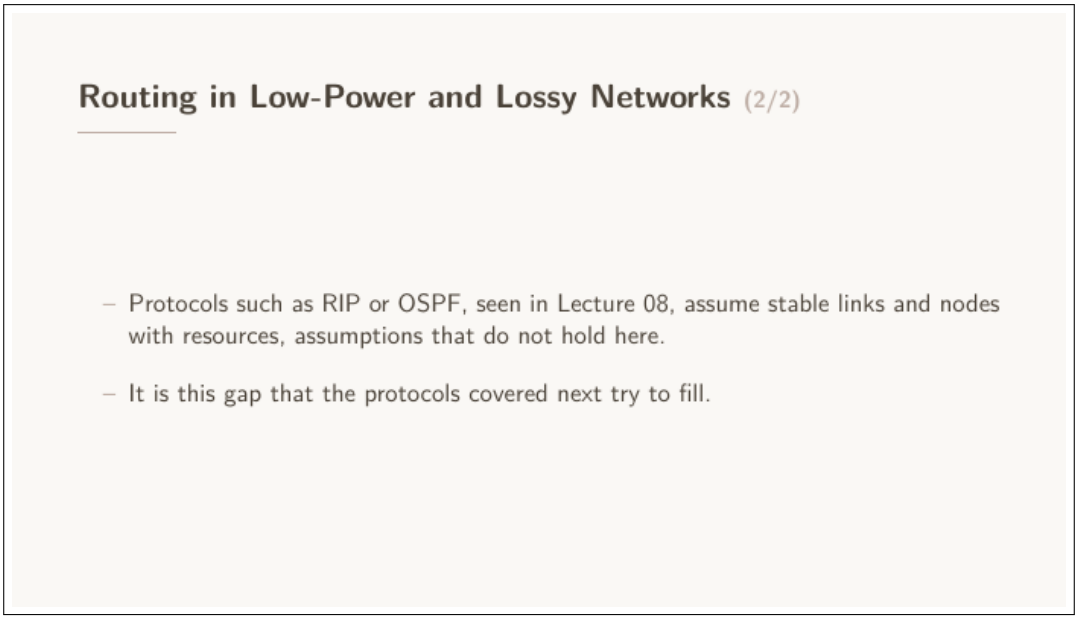

**The published standards.** RFC 7554 covers the use of TSCH, 8180 the minimal configuration and 8480 the 6top protocol. You should associate each number with its topic.

### Slide 26. Routing in Low-Power and Lossy Networks (1/2)



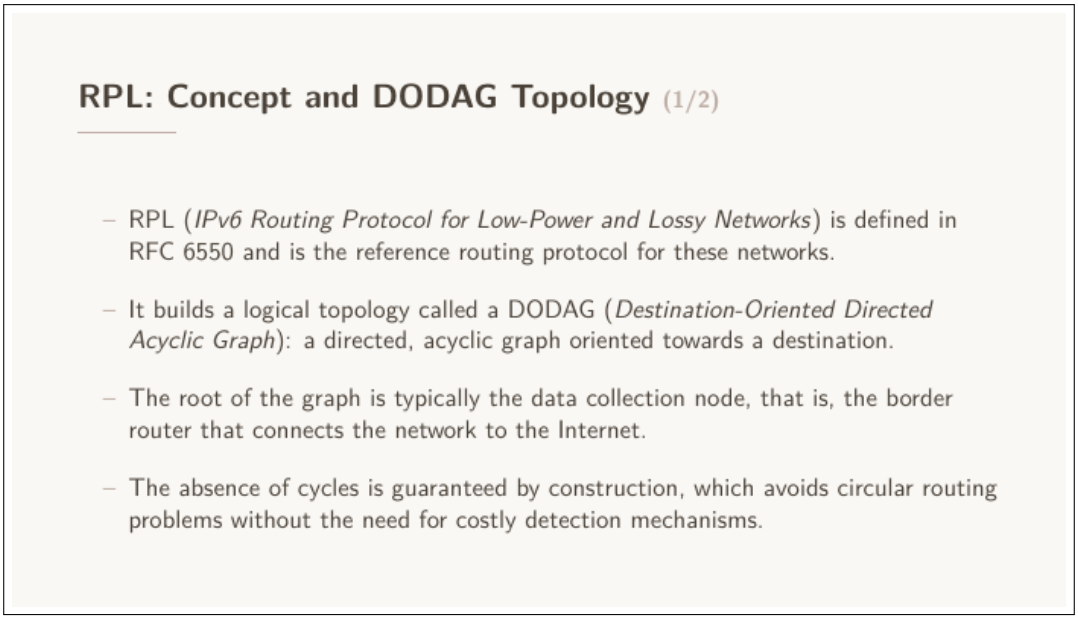

**Why classic protocols do not fit.** Limited memory and energy, unstable and asymmetric links, and traffic directed to a single collection point. Each characteristic invalidates an assumption of conventional protocols.

### Slide 27. Routing in Low-Power and Lossy Networks (2/2)



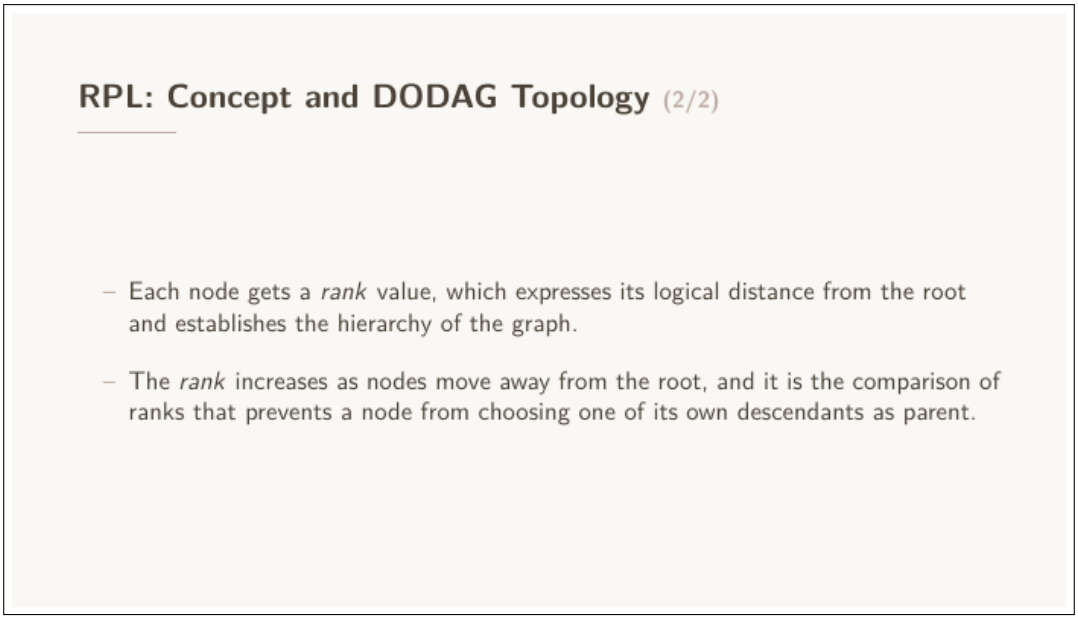

**A gap.** RIP and OSPF, studied in Lecture 08, assume stable links and nodes with resources. This gap is the one RPL and its alternatives fill.

### Slide 28. RPL: Concept and DODAG Topology (1/2)



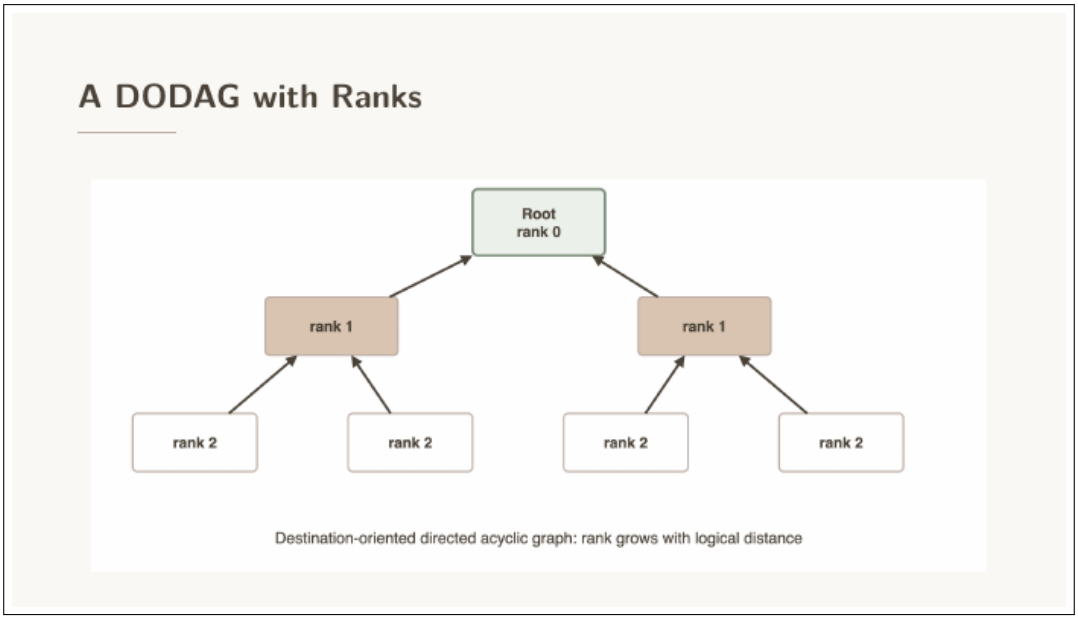

**The DODAG.** RPL organises the network in a directed acyclic graph oriented towards the root, typically the border router. Being acyclic by construction avoids circular routing without detection mechanisms.

### Slide 29. RPL: Concept and DODAG Topology (2/2)



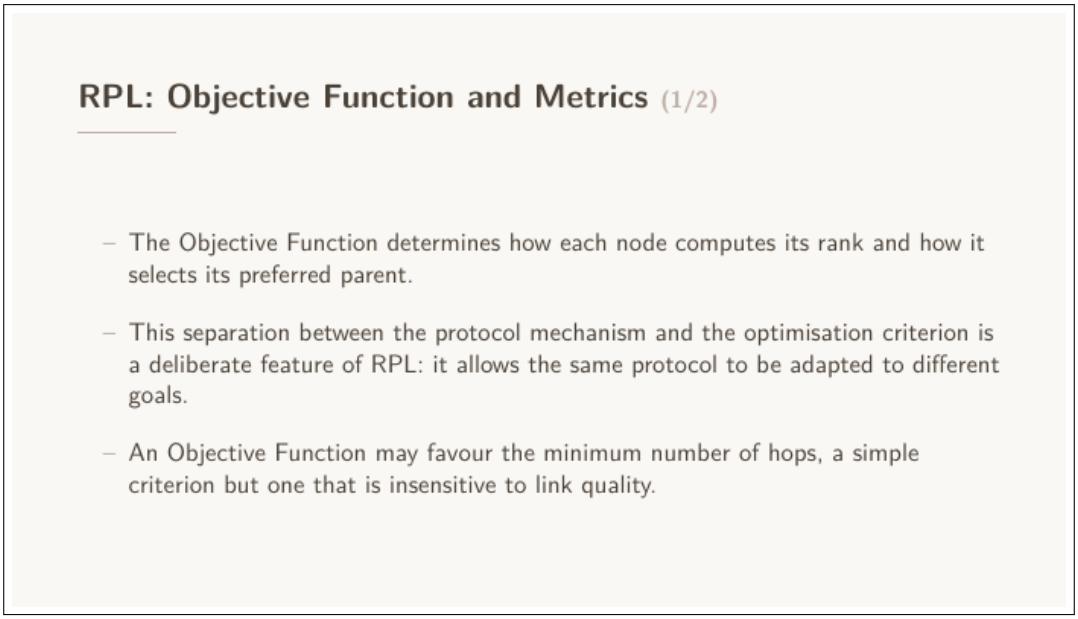

**The rank.** The rank expresses the logical distance from the root and grows with distance. It is the comparison of ranks that prevents a node from choosing a descendant as its parent.

### Slide 30. A DODAG with Ranks



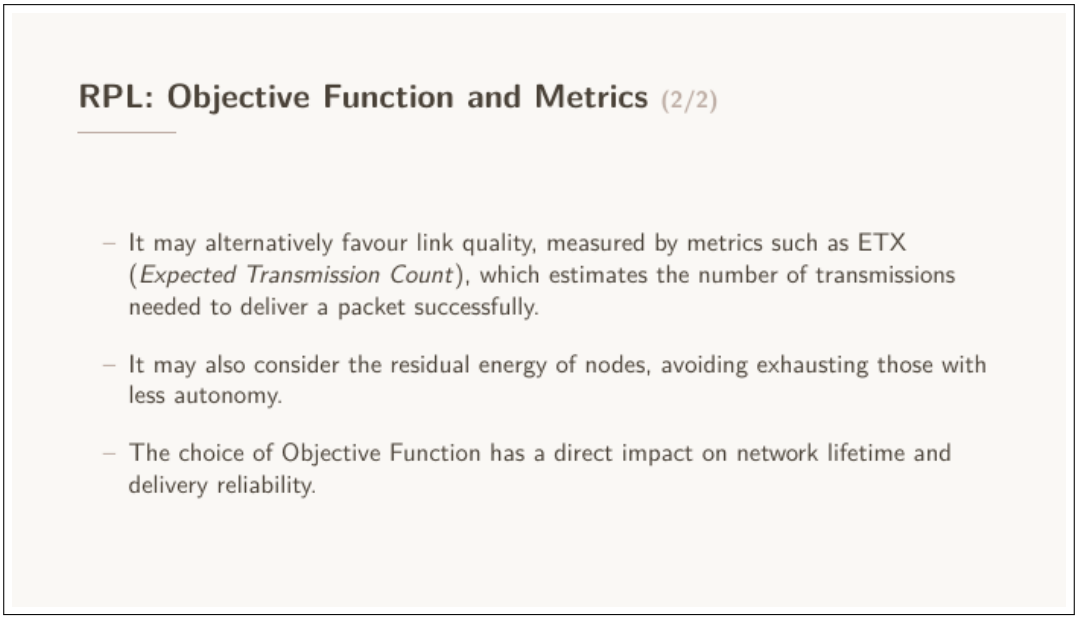

**A DODAG with ranks.** The figure shows the root with rank 0, the rank 1 nodes and the rank 2 nodes. Each node chooses as parent a node with a lower rank. Check: why are there no cycles?

### Slide 31. RPL: Objective Function and Metrics (1/2)



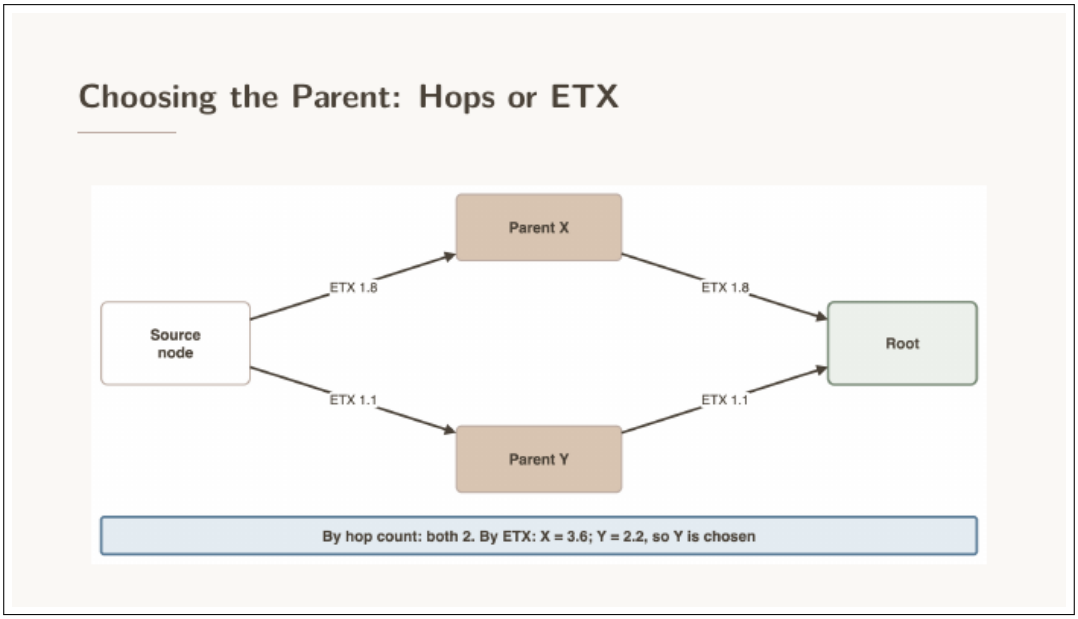

**The objective function.** The Objective Function says how rank is computed and how the parent is chosen. Separating the mechanism from the criterion allows the same protocol to be used with different goals. The minimum number of hops is simple but blind to quality.

### Slide 32. RPL: Objective Function and Metrics (2/2)



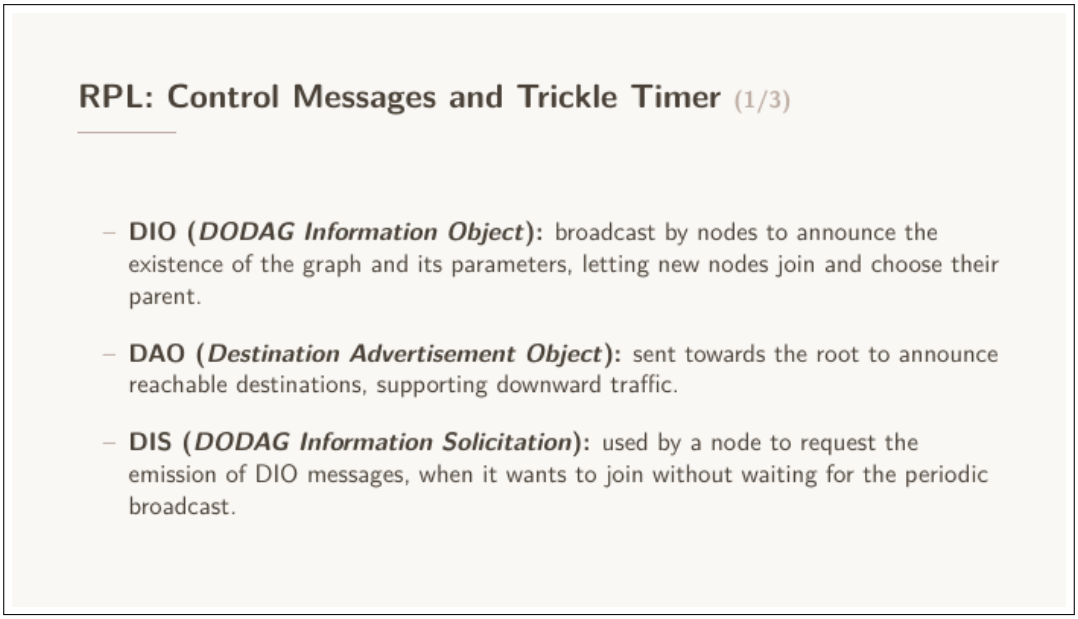

**ETX and energy.** ETX estimates how many transmissions are needed to deliver a packet. Residual energy avoids exhausting the weakest nodes. The choice of function has an impact on network lifetime and reliability.

## Slide 33. Choosing the Parent: Hops or ETX



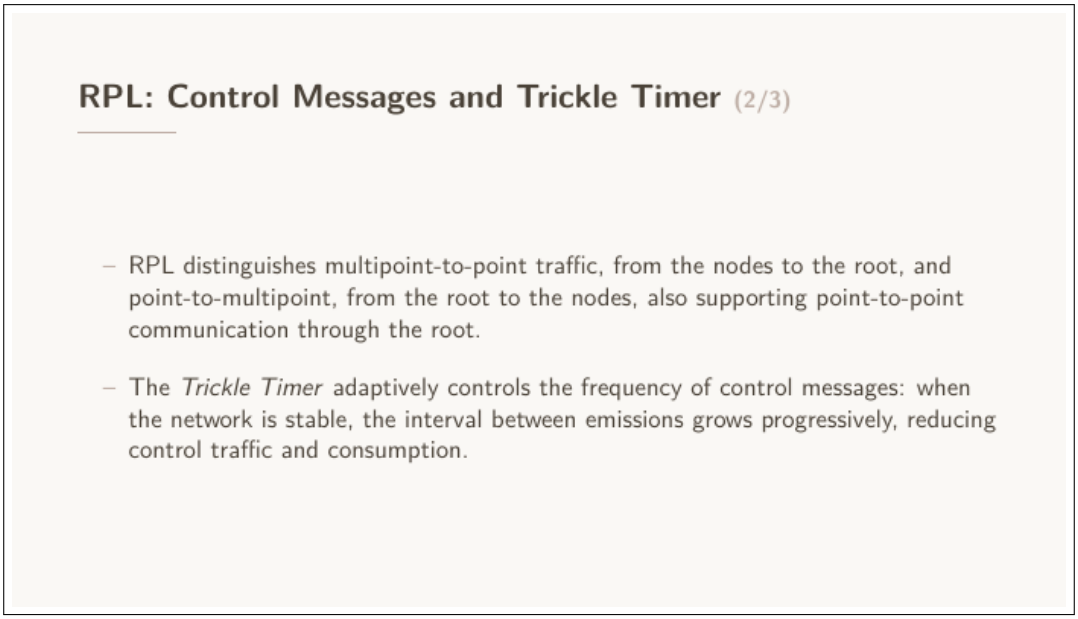

**Hops or ETX.** In the figure, both paths have two hops, but summing ETX the upper path is worth 3.6 and the lower one 2.2. The ETX-based function chooses parent Y, even though the hop count is the same.

### Slide 34. RPL: Control Messages and Trickle Timer (1/3)



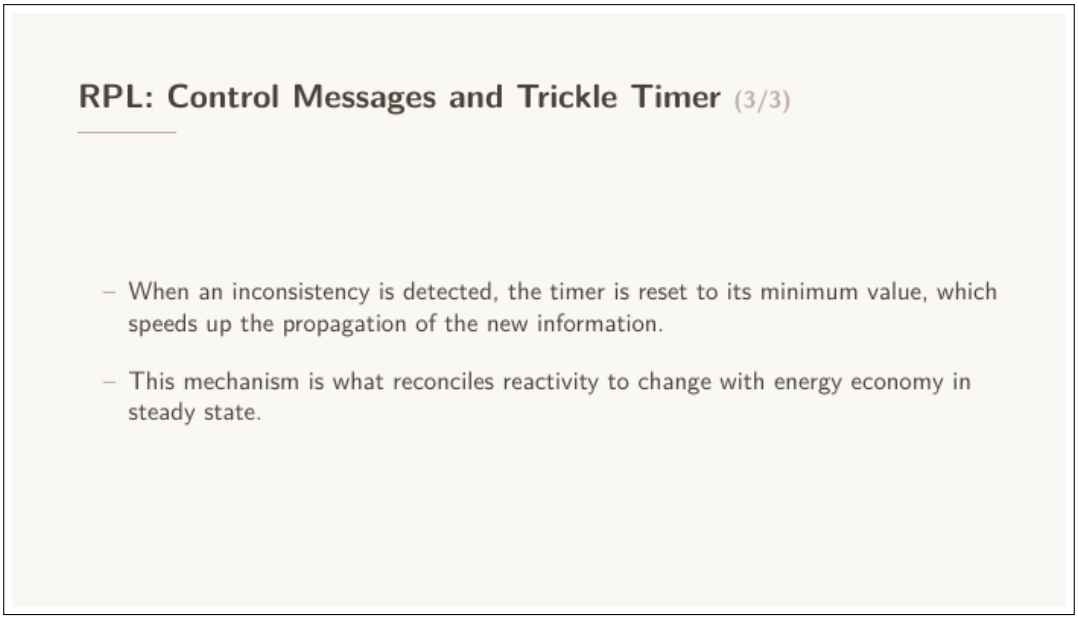

**DIO, DAO and DIS.** The DIO announces the graph, the DAO announces destinations towards the root and the DIS requests a DIO. You should know the direction of each: DIO goes down, DAO goes up.

### Slide 35. RPL: Control Messages and Trickle Timer (2/3)



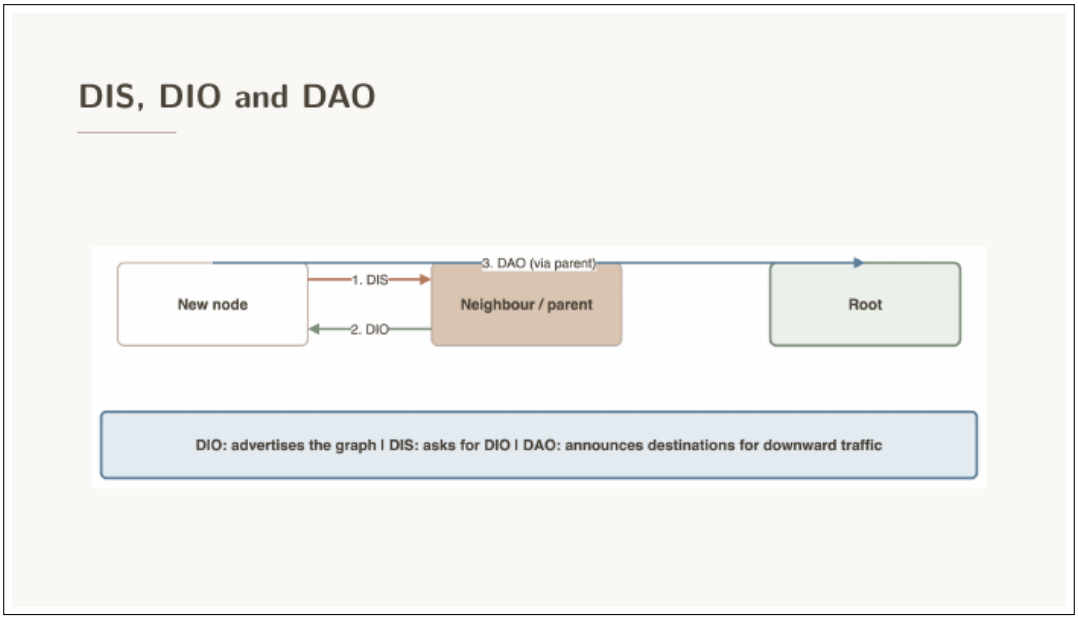

**Traffic types and Trickle.** There is traffic from the nodes to the root, from the root to the nodes and between peers through the root. The Trickle Timer lengthens the interval between DIOs when the network is stable, saving control traffic and energy.

### Slide 36. RPL: Control Messages and Trickle Timer (3/3)



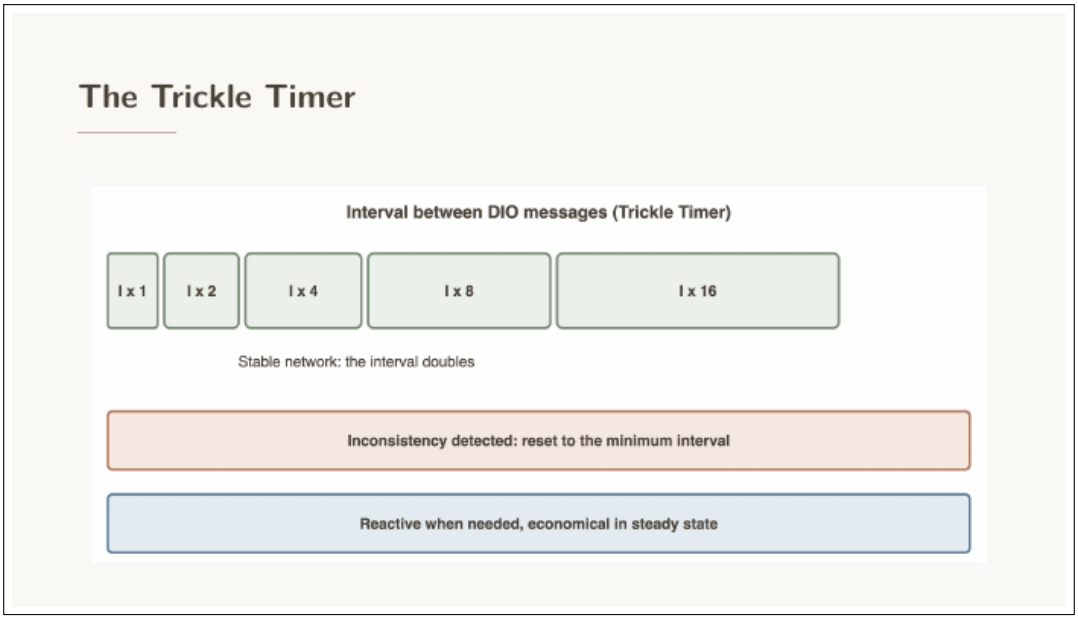

**Resetting the timer.** On detecting an inconsistency, the timer goes back to its minimum value and the new information spreads quickly. The mechanism reconciles reactivity with economy in steady state.

## Slide 37. DIS, DIO and DAO



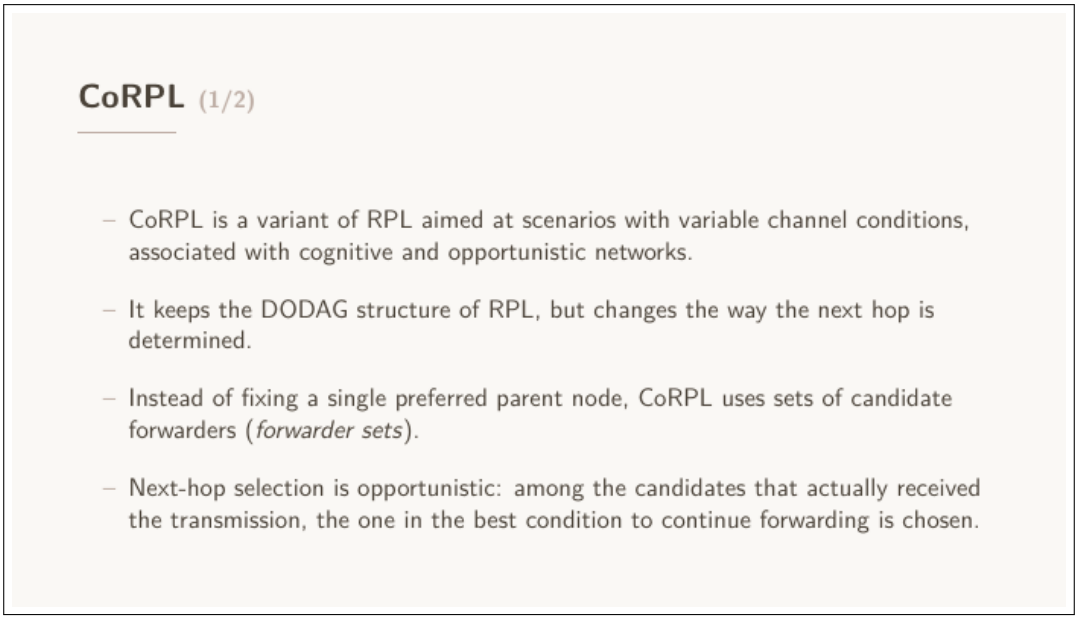

**The message exchange.** The figure shows a new node sending a DIS, receiving a DIO and announcing itself with a DAO. It is the typical sequence for joining an RPL network.

## Slide 38. The Trickle Timer



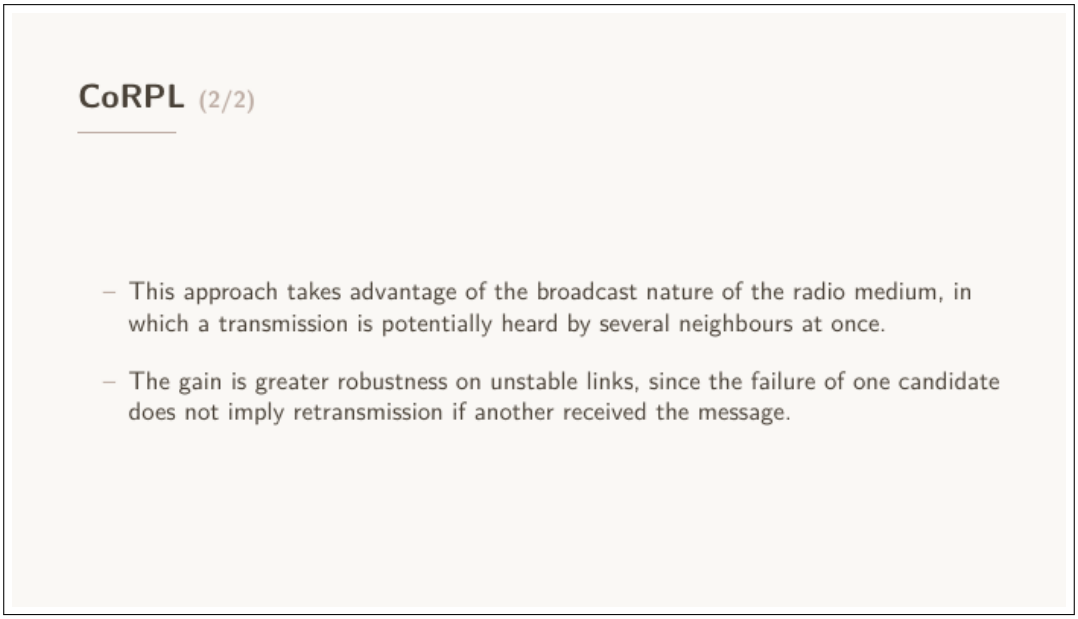

**Trickle in a picture.** The interval doubles while nothing changes and is reset to the minimum when an inconsistency is detected. That is why a stable network emits almost no signalling.

## Slide 39. CoRPL (1/2)



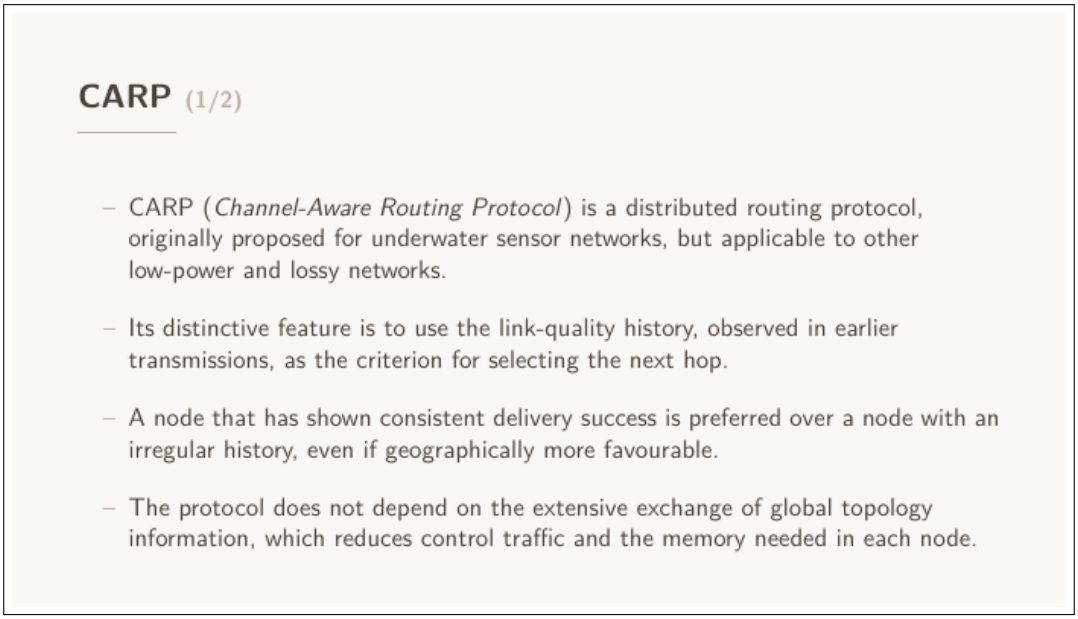

**CoRPL.** CoRPL keeps the DODAG but swaps the single parent for a set of candidates, opportunistically choosing the one best able to continue forwarding. It is designed for variable channels.

## Slide 40. CoRPL (2/2)



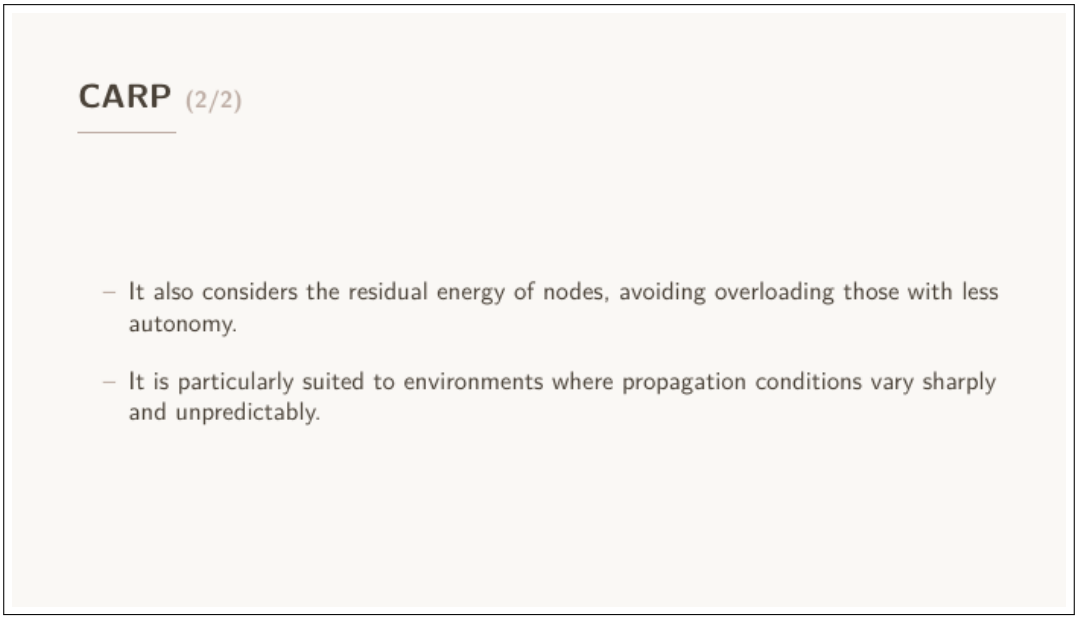

**Using radio broadcast.** A transmission may be heard by several neighbours. CoRPL takes advantage of this: if one candidate fails but another received, there is no retransmission. The gain is robustness on unstable links.

## Slide 41. CARP (1/2)



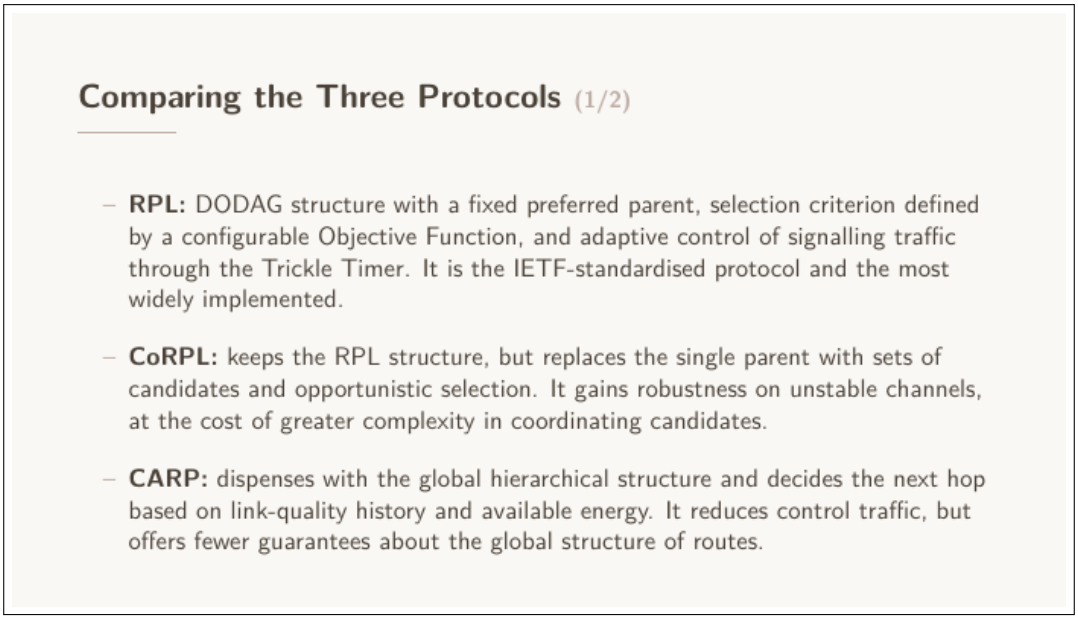

**CARP.** CARP uses link-quality history to choose the next hop, without exchanging global topology information. A neighbour with consistent success is preferred to a geographically better but irregular one.

## Slide 42. CARP (2/2)



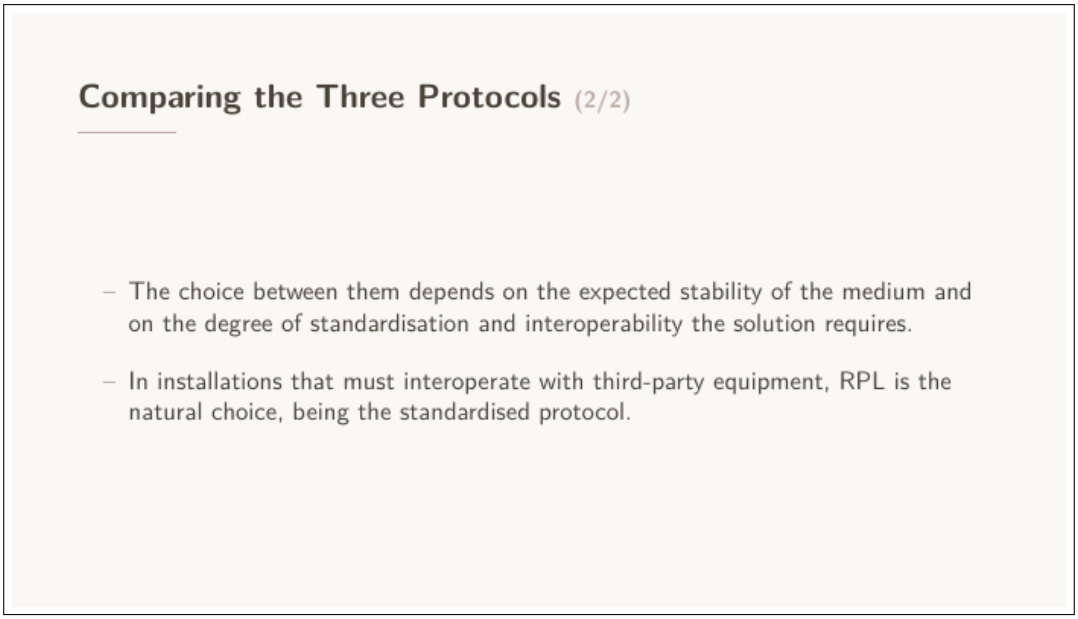

**Energy and unpredictable environments.** CARP considers residual energy and is suited to cases where propagation conditions vary a lot. It was born in underwater networks, but applies to LLNs in general.

### Slide 43. Comparing the Three Protocols (1/2)



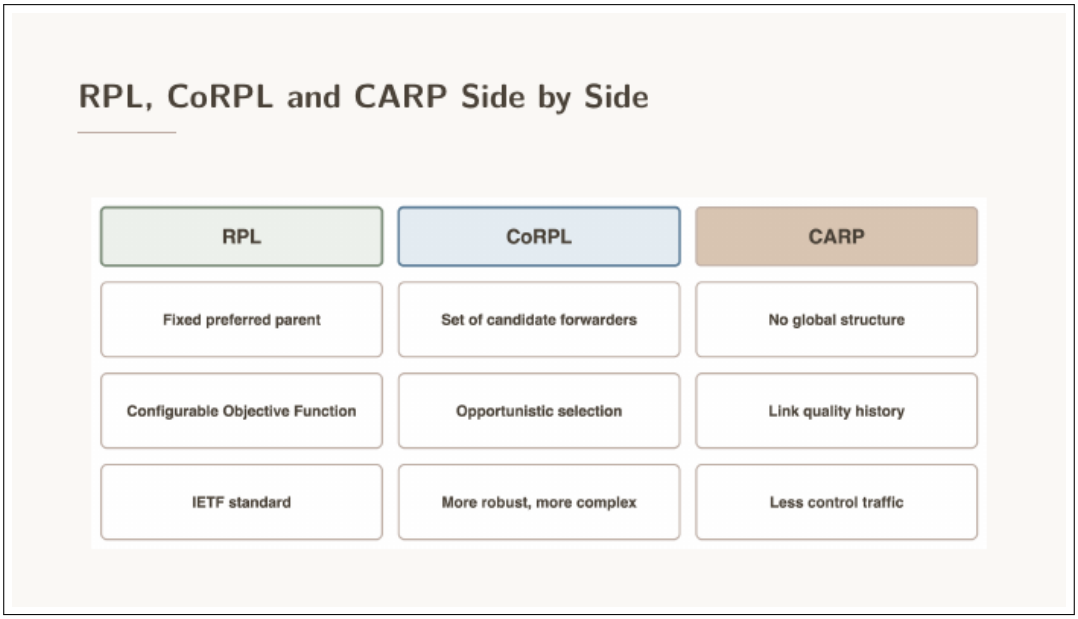

**Three protocols, three philosophies.** RPL has a fixed parent and a configurable function. CoRPL has candidates and opportunistic selection. CARP has history and no global structure. Each has an associated cost.

### Slide 44. Comparing the Three Protocols (2/2)



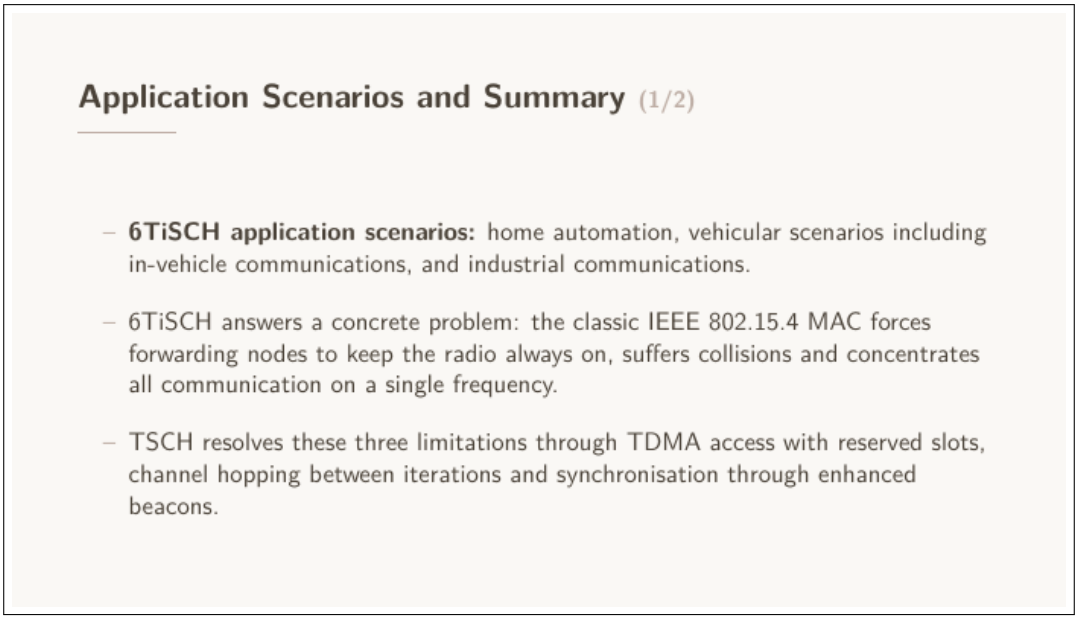

**When to choose each.** The choice depends on the stability of the medium and the interoperability required. For third-party equipment, RPL is the natural choice because it is the standardised one.

### Slide 45. RPL, CoRPL and CARP Side by Side



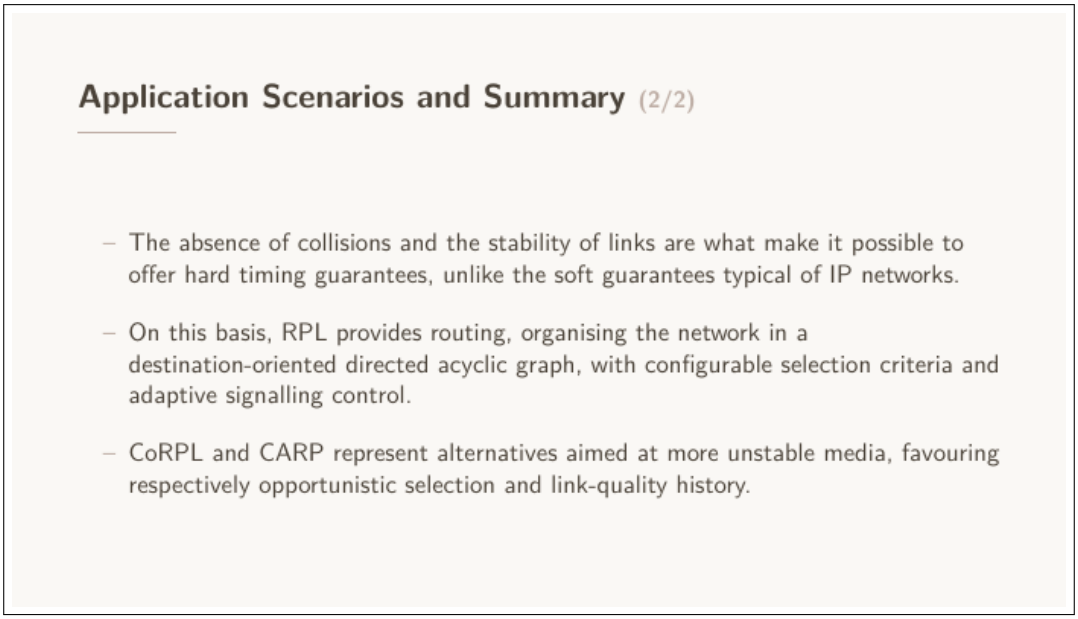

**Comparison in columns.** The figure summarises the three protocols in three columns with the same three rows: structure, criterion and cost. You should be able to rebuild it from memory.

## Slide 46. Application Scenarios and Summary (1/2)




**Scenarios and problem.** Home automation, vehicles and industry are the scenarios. The problem is that of the classic MAC: radio always on, collisions and a single frequency. TSCH solves all three with TDMA, channel hopping and beacons.

### Slide 47. Application Scenarios and Summary (2/2)




**The essentials of the lecture.** With no collisions and stable links it is possible to give hard timing guarantees. RPL provides routing, and CoRPL and CARP are alternatives for more unstable media.

## Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. What are the three defects of the classic 802.15.4 MAC and how does TSCH solve them?
2. What is a cell in a TSCH slotframe and how does channel hopping affect it?
3. What role does rank play in a DODAG, and why are there no cycles?
4. What is the difference between a hop-count-based and an ETX-based objective function?
5. What are the DIO, DAO and DIS messages for?
6. How does the Trickle Timer work and what advantage does it bring?
7. How do RPL, CoRPL and CARP differ in choosing the next hop?

## References


- T. Winter et al. *RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks.* RFC 6550, 2012.
- P. Thubert (ed.). *An Architecture for IPv6 over the TSCH mode of IEEE 802.15.4.* RFC 9030, 2021.
- D. Dujovne et al. *6TiSCH: Deterministic IP-Enabled Industrial Internet (of Things).* IEEE Communications Magazine, 2014.
- X. Vilajosana, K. Pister, T. Watteyne. *Minimal IPv6 over the TSCH Mode of IEEE 802.15.4e (6TiSCH) Configuration.* RFC 8180, 2017.